**JAVA LAB WEEK 10**

**Title: File Handling – Reading and Writing Files (Frequency Count of Patterns)**

**1. Objective / Aim**

**To write and execute a Java program to demonstrate file handling operations such as writing text into a file, reading the contents of a file, and finding the frequency of the patterns pe and pi in the given text.**

**2. System Requirements**

**Hardware Requirements:**

- **Computer or laptop**
- **Minimum 4 GB RAM**
- **Keyboard and monitor**

**Software Requirements:**

- **Operating System: Windows, Linux, or macOS**
- **Java Development Kit (JDK): JDK 8 or above**
- **IDE: Eclipse, IntelliJ IDEA, NetBeans, or Visual Studio Code**
- **Command Prompt or Terminal**

**3. Algorithm / Logic Description**

1. **Start the program.**
2. **Import the required Java I/O package.**
3. **Create a class named FileHandling.**
4. **Create a method named count() to count the occurrences of a given pattern in a string.**
5. **Create a FileWriter object to create the file named sample.txt.**
6. **Write the given text into the file using the write() method.**
7. **Close the FileWriter using the close() method.**
8. **Create a FileReader object to read the contents of the file.**
9. **Read the file character by character using the read() method.**
10. **Store the characters in a StringBuilder object.**
11. **Close the FileReader.**

12. Convert the complete text into lowercase using toLowerCase().

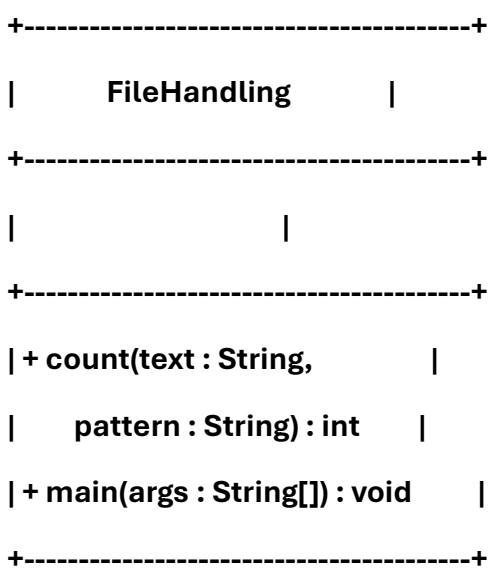
13. Call the count() method to find the occurrences of the pattern pe.

14. Call the count() method again to find the occurrences of the pattern pi.

15. Display the frequency count of both patterns.

16. Stop the program.

**4. UML Class Diagram**






**Description:**

- **FileHandling is the main user-defined class.**
- **count(String text, String pattern) is a method used to count the occurrences of a pattern.**
- **main() is the main method where file reading and writing operations are performed.**
- **FileWriter is used to write text into the file.**
- **FileReader is used to read data from the file.**
- **StringBuilder is used to store the characters read from the file.**
- **Objects are created for FileWriter, FileReader, and StringBuilder.**

**5. Source Code**

```
import java.io.*;

public class FileHandling {

  public static int count(String text, String pattern) {
    int count = 0;
    int index = 0;

    while ((index = text.indexOf(pattern, index)) != -1) {
      count++;
      index++;
    }

    return count;
  }

  public static void main(String[] args) throws IOException {

    // Writing text into the file
    FileWriter fw = new FileWriter("sample.txt");

    fw.write("Peter Piper picked a peck of pickled peppers\n");
    fw.write("A peck of pickled peppers Peter Piper picked\n");
    fw.write("If Peter Piper picked a peck of pickled peppers\n");
    fw.write("Where's the peck of pickled peppers Peter Piper picked?");

    fw.close();
```

```
        // Reading text from the file
        FileReader fr = new FileReader("sample.txt");

        StringBuilder data = new StringBuilder();
        int ch;

        while ((ch = fr.read()) != -1) {
            data.append((char) ch);
        }

        fr.close();

        // Converting text into lowercase
        String text = data.toString().toLowerCase();

        // Counting patterns
        int pe = count(text, "pe");
        int pi = count(text, "pi");

        // Displaying the result
        System.out.println("'pe' - no of occurrences - " + pe);
        System.out.println("'pi' - no of occurrences - " + pi);
    }
}
```

**6. Compilation and Execution Commands**

**Step 1: Save the program**

**Save the Java program with the filename FileHandling.java.**

**Step 2: Open Command Prompt or Terminal**

**Navigate to the folder where the Java file is saved.**

**Step 3: Compile the program**

**javac FileHandling.java**

**Step 4: Execute the program**

**java FileHandling**

**Step 5: File Creation**

**After successful execution, a file named sample.txt will be automatically created in the same working directory. The given text will be written into that file.**

**7. Output Screen**

**Sample Output:**

**'pe' - no of occurrences - 20**

**'pi' - no of occurrences - 12**

**Content Written into sample.txt:**

**Peter Piper picked a peck of pickled peppers**

**A peck of pickled peppers Peter Piper picked**

**If Peter Piper picked a peck of pickled peppers**

**Where's the peck of pickled peppers Peter Piper picked?**

**Explanation of Output:**

**The program reads the text from sample.txt and counts the occurrences of the given patterns.**

- **pe – The pattern occurs 20 times.**
- **pi – The pattern occurs 12 times.**

**The program converts the text into lowercase before counting, so uppercase and lowercase letters are treated equally.**

**Therefore, the final output is:**

- **pe = 20**
- **pi = 12**

**8. Result**

**Thus, the Java program to demonstrate file handling operations was successfully implemented and executed. The program creates a file named sample.txt, writes the given text into it, reads the file contents, and finds the frequency of the patterns pe and pi.**

**The experiment demonstrates file creation, writing data using FileWriter, reading data using FileReader, string handling using StringBuilder, and pattern counting using the indexOf() method.**